Replication files for "Public Pension Reforms and Retirement Decisions: Narrative Evidence and Aggregate Implications"

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Overview

The code in this replication package constructs the analysis file from several data sources (detailed below) using R, Stata, Matlab, and excel.

- The folder of Draft_Tables_Figs provides seven main files that run the code to generate all figures and tables (figure 1-16 and table 1) in the paper.
- The folder of Appendix_Tables_Figs provides eight files to generate all figures and tables (figure A1-C2 and table A2) in the appendix.

The replicator should expect the code to run less than 10 minutes.

Data Availability and Provenance Statements

Statement about Rights

- We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of Availability

- All data can be made publicly available.

Public use data collected by the authors

- The public pension reforms data used to support the findings of this study have been deposited in this archive [openICPSR-173182]. The data were collected by the authors and are available under a Creative Commons Non-commercial license.

  Datafile: `Draft_Tables_Figs\bz_narrative_data.xlsx`

Data that requires subscription to access

- This paper uses fiscal consolidation dataset from Guajardo, J., D. Leigh, and A. Pescatori (2014) "Expansionary Austerity? International Evidence," *Journal of the European Economic Association,* 12(4), 949-968. Dataset was accessed in 2020 and transcribed from the paper (Table 1). A copy of the data is provided as part of this archive. The paper can be downloaded from (with subscription): https://doi.org/10.1111/jeea.12083

  Datafile: `Draft_Tables_Figs\raw_data\ fiscal_consol.xlsx`

Public use data sourced from elsewhere and provided

- This paper uses labor market reform dataset by Romain A Duval, Davide Furceri, Bingjie Hu, João Tovar Jalles and Huy Nguyen (2018) "A Narrative Database of Major Labor and Product Market Reforms in Advanced Economies." Dataset was accessed in 2020 and a copy of the data is provided as part of this archive. We extracted the series on employment protection legislation reforms and unemployment benefit reforms for regular workers. The data are in the public domain and can be downloaded from: https://www.imf.org/en/Publications/WP/Issues/2018/01/25/A-Narrative-Database-of-Major-Labor-and-Product-Market-Reforms-in-Advanced-Economies-45585

  Datafile: `Draft_Tables_Figs\raw_data\Reforms_dataset_Online_Appendix.xlsx`

- This paper uses elderly population share from the Organisation for Economic Co-operation and Development Database. 1960-2018a. "Elderly Population Share." Dataset was accessed in 2019 and a copy of the data is provided as part of this archive. The data are in the public domain and can be downloaded from: https://data.oecd.org/pop/elderly-population.htm#indicator-chart

  Datafile: Datafile: `Draft_Tables_Figs\raw_data\epop.xlsx`

- This paper uses labor force participation rates from the Organisation for Economic Co-operation and Development Database. 1960-2018b. "Labor Force Participation Rate." Aggregate. Accessed via https://data.oecd.org/emp/labour-force-participation-rate.htm
  By age and gender. Accessed via
  https://stats.oecd.org/Index.aspx?DataSetCode=lfs_sexage_i_r
  Datasets were accessed in 2019 and copies of the data are provided as part of this archive.

  Datafiles:

```
Aggregate LFPR: Draft_Tables_Figs\raw_data\lf.xlsx
LFPR 55 to 64: Draft_Tables_Figs\raw_data\lf_55to59.xlsx
LFPR 55 to 64: Draft_Tables_Figs\raw_data\lf_55to64.csv
LFPR 60 to 64: Draft_Tables_Figs\raw_data\lf_60to64.xlsx
LFPR 64 to 69: Draft_Tables_Figs\raw_data\lf_64to69.xlsx
LFPR 20 to 49 constructed as population weighted average using the two
files (details in step2_construct_bz_data.do):
Draft_Tables_Figs\raw_data\lf_20to49.xslx
Draft_Tables_Figs\raw_data\age_pop.csv
Male LFPR 55 to 59: Draft_Tables_Figs\raw_data\men_5559.xlsx
Male LFPR 55 to 64: Draft_Tables_Figs\raw_data\men5564.xlsx
Male LFPR 60 to 64: Draft_Tables_Figs\raw_data\men6064.xlsx
Female LFPR 55 to 59: Draft_Tables_Figs\raw_data\women_5559.xlsx
```

Female LFPR 55 to 64: Draft_Tables_Figs\raw_data\women5564.xlsx
Female LFPR 60 to 64: Draft_Tables_Figs\raw_data\women6064.xlsx

- This paper uses government spending data from the Organisation for Economic Co-operation and Development Database. 1960-2018c. "Government Spending." Accessed via https://data.oecd.org/gga/general-government-spending.htm
  Dataset was accessed in 2019 and a copy of the data is provided as part of this archive.

  Datafile: Draft_Tables_Figs\raw_data\gov_y.xlsx

- This paper uses Gross Domestic Product from the Organisation for Economic Co-operation and Development Database. 1960-2018d. "Gross Domestic Product." Accessed via https://data.oecd.org/gdp/gross-domestic-product-gdp.htm
  Dataset was accessed in 2019 and a copy of the data is provided as part of this archive.

  Datafile: Draft_Tables_Figs\raw_data\ngdp_y.xlsx

- This paper uses consumer price index from the Organisation for Economic Co-operation and Development Database. 1960-2018e. "Consumer Price Index." Accessed via https://data.oecd.org/price/inflation-cpi.htm
  Dataset was accessed in 2019 and a copy of the data is provided as part of this archive.

  Datafile: Draft_Tables_Figs\raw_data\cpi_index.xlsx

- This paper uses tax revenues from Organisation for Economic Co-operation and Development Database. 1960-2018f. "Tax revenues." Accessed via https://data.oecd.org/tax/tax-revenue.htm
  Dataset was accessed in 2019 and a copy of the data is provided as part of this archive.

  Datafile: Draft_Tables_Figs\raw_data\taxrev,xlsx

- This paper uses pension spending from Organisation for Economic Co-operation and Development Database. 1980-2018. "Old Age Pension Spending as percentage of GDP." Accessed via https://stats.oecd.org/Index.aspx?datasetcode=SOCX_AGG
  Dataset was accessed in 2019 and a copy of the data is provided as part of this archive.

  Datafile: Draft_Tables_Figs\raw_data\pen_oldage.xlsx

- This paper uses unemployment rate from Organisation for Economic Co-operation and Development Database. 1980-2018. "Unemployment rate." Accessed via https://data.oecd.org/unemp/unemployment-rate.htm
  Dataset was accessed in 2019 and a copy of the data is provided as part of this archive.

  Datafile: `Draft_Tables_Figs\raw_data\uemp_y.xlsx`

- This paper uses life expectancy from World Bank. 1978-2018. "Life expectancy." Accessed via https://data.worldbank.org/indicator/SP.DYN.LE00.IN
  Dataset was access edin 2019 and a copy of the data is provided as part of this archive.

  Datafile: `Draft_Tables_Figs\raw_data\life_expectancy.xlsx`

Dataset list

Data file	Source	Notes	Provided
`bz_narrative_data.xlsx`	Constructed		Yes
`Dummy.csv`	Constructed		Yes
`raw_data\Reforms_dataset_Online_Appendix.xlsx`	Romain et al. (2018)		Yes
`raw_data\ fiscal_consol.xlsx`	Guajardo et al. (2014)		Yes
`raw_data\epop.xlsx`	OECD	As per terms of use	Yes
`raw_data\lf.xlsx`	OECD	As per terms of use	Yes
`raw_data\lf_55to64.xlsx`	OECD	As per terms of use	Yes
`raw_data\lf_60to64.xlsx`	OECD	As per terms of use	Yes

raw_data\lf_64to69.xlsx	OECD	As per terms of use	Yes
raw_data\lf_20to49.xlsx	OECD	As per terms of use	Yes
raw_data\age_pop.csv	OECD	As per terms of use	Yes
raw_data\men_5559.xlsx	OECD	As per terms of use	Yes
raw_data\men_5564.xlsx	OECD	As per terms of use	Yes
raw_data\men_6064.xlsx	OECD	As per terms of use	Yes
raw_data\women_5559.xlsx	OECD	As per terms of use	Yes
raw_data\women_5564.xlsx	OECD	As per terms of use	Yes
raw_data\women_6064.xlsx	OECD	As per terms of use	Yes
raw_data\gov_y.xlsx	OECD	As per terms of use	Yes
raw_data\ngdp_y,xlsx	OECD	As per terms of use	Yes
raw_data\cpi_index.xlsx	OECD	As per terms of use	Yes
raw_data\taxrev,xlsx	OECD	As per terms of use	Yes
raw_data\pen_oldage,xlsx	OECD	As per terms of use	Yes

raw_data\life_expectancy,xlsx	World Bank	As per terms of use	Yes
raw_data\uemp_y.xlsx	OECD	As per terms of use	Yes

Computational requirements

Software Requirements

- Stata (code needed to be run with version 17)
- Matlab (code was run with Matlab Release 2020a)
- R 1.3.959
    1. `readxl`
    2. `tidyverse`

Memory and Runtime Requirements

The code was last run on an Intel-based laptop with Windows 10 Enterprise 21H2 with 32 GB of RAM. Computation took less than 10 minutes.

Description of programs/code

- Programs and files in ./Draft_Tables_Figs generate all figures and tables (figure 1-16 and table 1) in the paper.
    1. **bz_narrative_data.xlsx**: This file documents pension policy changes in 10 countries between 1962 and 2019, see section 3 in the paper for more methodology details. It includes a "ReadMe" tab, one tab for each country with pension policy changes, as well as a tab of "All_countries" that consolidates pension policy changes across 10 countries.

    2. **Step1_bz_dummies.R**: This file loads in the narrative data file ("All_countries" in bz_narrative_data.xlsx) and generates the public pension reform dummy file (Dummy.csv) with the following dummy series across 10 countries over time:
       Reform_index Reform_delay Major_nodelay Major_delay
       Major_delay_age Major_delay_contribution Major_delay_other
       Major_delay_short Major_delay_long Major_delay_age_short
       Major_delay_contribution_short Major_delay_age_long

Major_delay_contribution_long

3. **Step2_construct_bz_data.do:** This file loads in pension reform dummy file (Dummy.csv) as well as other country-specific data from the folder of raw_data (details see the section of Data Availability and Provenance Statements). It then generates bz_oecd_data.csv, the data input file for STATA to be used in empirical analysis of the paper.

4. **subfolder "step3_figure_1"**: replicate figure 1.
data_figure_1.xlsx includes the underlying data for figure 1 as well as a "ReadMe" tab for data source information. Run figure_1.R file to reproduce figure 1. The code was run on RStudio v1.3.959.

5. **step4_figures_2to5.xlsx**: replicate figures 2 to 5.
This file includes the underlying data for figures 2 to 5. It also includes a "ReadMe" tab that explains the steps to construct those figures.

6. **Step5_figure_6.xlsx**: replicate figure 6.
This file includes the underlying data for figure 6. It also includes a "ReadMe" tab that explains the steps to construct the figure.

7. **Step6_main_figures_7to16_table_1**: replicate figures 7 to 16 as well as table 1.
Run the file to produce the figures. The files use bz_oecd_data.csv and were run on Stata/SE 17.0. It calls figures_7_8.do, figures_9_10_12.do, figures_11_13.do, figures_14_16.do, figure_15.do, and table_1.do. The figures are stored in the folder of "graphs". The description and legend in each figure correspond to the figure description within the paper.

- Programs and files in ./Appendix_Tables_Figs generate all figures and tables (figure A1-C2 and table A2) in the online appendix.

1. **Figure_A1.xlsx**: replicate figures A1 (a) and (b) in the online appendix
This file includes the underlying data for figures A1 (a) and (b). It also includes a "ReadMe" tab that explains the steps to construct the figures as well as a "OECD_source" tab that provides underlying raw data.

2. **Figure_A2_A3.xlsx**: replicate figures A2 and A3 in the online appendix.
This file includes the underlying data for figures A2 and A3. It also includes a "ReadMe" tab that explains the steps to construct the figures as well as a "OECD_source" tab that provides underlying raw data.

3. **Figure_A4.xlsx**: replicate figure A4 in the online appendix.
This file includes the underlying data for figure A4. It also includes a "ReadMe" tab that explains the steps to construct the figure.

4. **Subfolder of ./figure_A5**: replicate figure A5 in the online appendix.

- Run **distribution_good_badtimes.m** to generate and display the underlying data. It also plots the bar chart of figure A5.
- **figure_A5.xlsx** plot a polished and formatted figure A5 in excel. This step is optional.

5. **Figure_A6.do**: replicate figures A6(a) and (b) in the online appendix.
   Run the file to produce the figures. The files use bz_oecd_data.csv and were run on Stata/SE 17.0. The figures are stored in the subfolder of "graphs".
6. **Figure_B1_B2_B3_C2.xlsx**: provides the data sources/links for the snapshots in figure B1, B2, B3, and C2.
7. **Figure_C1.xlsx**: replicate figure C1 in the online appendix.
   This file shows how to construct figure C1.
8. **Table_A2.xlsx**: replicate figure A2 in the online appendix.
   This file includes the underlying data for Table A2. It also includes a "ReadMe" tab that explains the steps to construct the figure.

Instructions to Replicators

- `Draft_Tables_Figs/ Step1_bz_dummies.R`: will create the public pension reform dummy file (Dummy.csv).
  1. The program was last run in October 2022.
- `Draft_Tables_Figs/Step2_construct_bz_data.do:` will create bz_oecd_data by loading in pension reform dummy file as well as other datasets in ./raw data.
- `Draft_Tables_Figs/Step3_figure_1:`
  1. Figure_1.R will create figure 1.
  2. The program was last run in October 2022.
- `Draft_Tables_Figs/ step4_figures_2to5.xlsx:` will create figures 2 to 5.
- `Draft_Tables_Figs/ step5_figure_6.xlsx:` will create figure 6.
- `Draft_Tables_Figs/ step6_main_figures_7to16_table_1.do:` will create figures 7 to 16.
  1. It calls figures_7_8.do, figures_9_10_12.do, figures_11_13.do, figures_14_16.do, figure_15.do, and table_1.do.
  2. Figures are saved in the subfolder of ./graphs.

3. The program was last run in October 2022.

- Appendix_Tables_Figs/figure_A1.xlsx: will create figures A1 (a) and (b).
- Appendix_Tables_Figs/figure_A2_A3.xlsx: will create figures A2 and A3.
- Appendix_Tables_Figs/figure_A4.xlsx: will create figures A4.
- Appendix_Tables_Figs/figure_A5:
    1. Run distribution_good_badtimes.m to create figure A5.
    2. The program was last run in October 2022.
- Appendix_Tables_Figs/figure_A6.do: will create figure 6 (A) and (B).
    1. Figures are saved in the subfolder of ./graphs.
    2. The program was last run in October 2022.
- Appendix_Tables_Figs/figure_B1_B2_B3_C2.xlsx: will create figures B1, B2, B3, and C2.
- Appendix_Tables_Figs/figure_C1.xlsx: will create figure C1.
- Appendix_Tables_Figs/table_A2.xlsx: will create table A2.

List of tables and programs

The provided code reproduces:

- All numbers provided in text in the paper

Figure/Table #	Program	Line number	Output file	Note
Figure 1	Draft_Tables_Figs/Step3_figure_1/Figure_1.do	29	Figure_1.png	
Figure 2	Draft_Tables_Figs/step4_figures_2to5.xlsx	7-9	N/A	In tab fig_2
Figure 3	Draft_Tables_Figs/step4_figures_2to5.xlsx	5-11	N/A	In tab fig_3
Figure 4	Draft_Tables_Figs/step4_figures_2to5.xlsx	3-12	N/A	In tab fig_4
Figure 5	Draft_Tables_Figs/step4_figures_2to5.xlsx	Col G-J	N/A	In tab fig_5
Figure 6	Draft_Tables_Figs/step5_figure_6.xlsx	Col G-J	N/A	
Figure 7a and b	Draft_Tables_Figs/ figures_7_8.do	197	Figure7a.pdf Figure7b.pdf	

Figure 8	Draft_Tables_Figs/ figures_7_8.do	197	Figure8.pdf	
Figure 9	Draft_Tables_Figs/ figures_9_10_12.do	129	Figure9.pdf	
Figure 10	Draft_Tables_Figs/ figures_9_10_12.do	320	Figure10.pdf	
Figure 11	Draft_Tables_Figs/ figures_11_13.do	149	Figure11.pdf	
Figure 12	Draft_Tables_Figs/ figures_9_10_12.do	512	Figure12.pdf	
Figure 13	Draft_Tables_Figs/ figures_11_13.do	433	Figure13.pdf	
Figure 14	Draft_Tables_Figs/ figures_14_16.do	141	Figure14.pdf	
Figure 15a and b	Draft_Tables_Figs/ figures_15.do	174	Figure15a.pdf; figure15b.pdf	
Figure 16a and b	Draft_Tables_Figs/ figures_14_16.do	519	Figure16a.pdf; Figure16b.pdf	
Table 1	Draft_Tables_Figs/ table_1.do	80	N/A	
Figure A1a and b	Appendix_Tables_Figs/figure_A1.xlsx	Col I-J Col A-E	N/A	
Figure A2	Appendix_Tables_Figs/figure_A2_A3.xlsx	Col A-C	N/A	In tab fig A2
Figure A3	Appendix_Tables_Figs/figure_A2_A3.xlsx	Col D-E	N/A	In tab fig A3
Figure A4	Appendix_Tables_Figs/figure_A4.xlsx	Col I-L	N/A	
Figure A5	Appendix_Tables_Figs/figure_A5/distribution_good_badtimes.m	71	N/A	
Figure A6a and b	Appendix_Tables_Figs/figure_A6.do	194	FigureA6a.pdf FigureA6b.pdf	
Figure B1	Appendix_Tables_Figs/figure_B1_B2_B3_C2.xlsx	Col A-G	N/A	In tab fig B1
Figure B2	Appendix_Tables_Figs/figure_B1_B2_B3_C2.xlsx	Col A-O	N/A	In tab fig B2
Figure B3	Appendix_Tables_Figs/figure_B1_B2_B3_C2.xlsx	Col A-H	N/A	In tab fig B3
Figure C1	Appendix_Tables_Figs/figure_C1.xlsx	Col A-E	N/A	
Figure C2	Appendix_Tables_Figs/figure_B1_B2_B3_C2.xlsx	Col A-H	N/A	In tab fig C2

References

Romain A Duval, Davide Furceri, Bingjie Hu, João Tovar Jalles and Huy Nguyen (2018) "A Narrative Database of Major Labor and Product Market Reforms in Advanced Economies." Dataset accessed from: https://www.imf.org/en/Publications/WP/Issues/2018/01/25/A-Narrative-Database-of-Major-Labor-and-Product-Market-Reforms-in-Advanced-Economies-45585

Guajardo, J., D. Leigh, and A. Pescatori (2014) "Expansionary Austerity? International Evidence," *Journal of the European Economic Association,* 12(4), 949-968. Dataset accessed from (Table 1): https://doi.org/10.1111/jeea.12083

Organisation for Economic Co-operation and Development Database. 1960-2018a. "Elderly Population Share." Dataset accessed from: https://data.oecd.org/pop/elderly-population.htm#indicator-chart

Organisation for Economic Co-operation and Development Database. 1960-2018b. "Labor Force Participation Rate." Aggregate. Accessed via https://data.oecd.org/emp/labour-force-participation-rate.htm
By age and gender. Accessed via
https://stats.oecd.org/Index.aspx?DataSetCode=lfs_sexage_i_r

Organisation for Economic Co-operation and Development Database. 1960-2018c. "Government Spending." Accessed via https://data.oecd.org/gga/general-government-spending.htm

Organisation for Economic Co-operation and Development Database. 1960-2018d. "Gross Domestic Product." Accessed via https://data.oecd.org/gdp/gross-domestic-product-gdp.htm

Organisation for Economic Co-operation and Development Database. 1960-2018e. "Consumer Price Index." Accessed via https://data.oecd.org/price/inflation-cpi.htm

Organisation for Economic Co-operation and Development Database. 1960-2018f. "Tax revenues." Accessed via https://data.oecd.org/tax/tax-revenue.htm

Organisation for Economic Co-operation and Development Database. 1980-2018. "Old Age Pension Spending as percentage of GDP." Accessed via
https://stats.oecd.org/Index.aspx?datasetcode=SOCX_AGG

Organisation for Economic Co-operation and Development Database. 1980-2018. "Unemployment rate." Accessed via
https://data.oecd.org/unemp/unemployment-rate.htm

World Bank. 1978-2018. “Life expectancy.” Accessed via https://data.worldbank.org/indicator/SP.DYN.LE00.IN